



iCET Certified Master

COURSEWORK

Module	Standalone Application Development
Coursework Title	Day of the Week Finder
Coursework Type	Course Work

Day of the Week Finder Using JavaScript

Overview

This assignment focuses on building a simple Day of the Week Finder web application using HTML, CSS, and JavaScript. The program will take user input for a specific date (year, month, and day) and calculate the corresponding day of the week using the Gregorian calendar formula. Students must also design a clean and user-friendly interface using CSS.

Learning Outcomes

- By completing this assignment, students will be able to:
- Handle user inputs and perform arithmetic calculations using JavaScript.
- Apply mathematical and logical operations to solve real-world problems.
- Use JavaScript's built-in functions like `Math.floor()` for integer division.
- Dynamically display and style results using HTML, CSS, and JavaScript.
- Design a simple, readable, and user friendly web layout using CSS.

Description

Students must create a web page that:

- Accepts user inputs for Year, Month, and Day through three input fields.
- Uses JavaScript to:
 - Read input values.
 - Compute the day of the week using the provided Gregorian formulas.
 - Map the result to the correct weekday (Sunday–Saturday).
 - Display the result dynamically when the user clicks a “Find Day” button.
- Includes CSS styling to improve visual appearance, including:
 - Centered layout
 - Proper spacing and labels for input fields
 - A styled button and result area

Formula to Use

$$y_o = y - ((14 - m) / 12)$$

$$x = y_o + y_o/4 - y_o/100 + y_o/400$$

$$m_o = m + 12 \times ((14 - m) / 12) - 2$$

$$d_o = (d + x + (31 \times m_o) / 12) \% 7$$

Note:

Use Math.floor() to perform integer division (rounding down to the nearest whole number). Convert user inputs (from text fields) to numbers using Number() or parseInt() before performing calculations.

Example**Input:**

Year = 2000, Month = 2, Day = 14

Calculation:

$$y_o = 2000 - ((14 - 2) / 12) = 1999$$

$$x = 1999 + 499 - 19 + 4 = 2483$$

$$m_o = 2 + 12 \times ((14 - 2) / 12) - 2 = 12$$

$$d_o = (14 + 2483 + (31 \times 12) / 12) \% 7 = 1$$

Output:

Day of the week is **Monday**

Submission Requirements

- Submit the project folder containing your HTML, CSS, and JavaScript files as a ZIP file.
- The interface must display results directly on the web page.
- Include proper comments in your JavaScript and CSS files explaining key sections.

Deadline

- Refer to the LMS for the official deadline date.

Late submissions will negatively impact your academic progress and may affect your final evaluation.